

SAFETY DATA SHEET (GHS)

Cumene Hydroperoxide (CHP) | CAS: 80-15-9 | Formula: C₉H₁₂O₂

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Cumene hydroperoxide, CAS 80-15-9 (<https://echa.europa.eu/>)
- PubChem CID 6629 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/6629>
- NTP/NIEHS, 'Cumene Hydroperoxide [80-15-9] Review of Toxicological Literature' (<https://ntp.niehs.nih.gov/>)
- Sigma-Aldrich SDS, Cumene hydroperoxide technical ~80%, product 247502 ([sigmaaldrich.com](https://www.sigmaaldrich.com))
- Thermal decomposition / SADT studies of 88 wt% CHP (Process Safety & Environmental Protection literature, ScienceDirect)
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Cumene Hydroperoxide (CHP)

CAS Number: 80-15-9

Chemical formula: C₉H₁₂O₂

Molecular weight: 152.19 g/mol

Synonyms: alpha,alpha-Dimethylbenzyl hydroperoxide; Isopropylbenzene hydroperoxide; Cumyl hydroperoxide; CHP

Commercial form note: Supplied industrially as ~80 wt% solution in cumene; properties below refer to this technical grade unless stated otherwise

Intended use: Process intermediate — product of cumene oxidation (OXI section); feed to CDN decomposition reactors.
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Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Organic Peroxide, Type E — H242; Flammable Liquid, Category 4 — H227 (combustible); Acute Tox. 4 (oral) — H302; Acute Tox. 4 (dermal) — H312; Acute Tox. 3 (inhalation) — H331; Skin Corr. 1B — H314 (also covers serious eye damage); STOT RE 2 — H373; Aquatic Acute 2 / Aquatic Chronic 2 — H401/H411

Signal word: Danger

Hazard statements (H-codes): H242 (Heating may cause a fire); H302 (Harmful if swallowed); H312 (Harmful in contact with skin); H314 (Causes severe skin burns and eye damage); H331 (Toxic if inhaled); H373 (May cause damage to organs — kidney, liver — through prolonged or repeated exposure); H411 (Toxic to aquatic life with long lasting effects)

Precautionary statements (P-codes): P210, P220, P234, P280, P283, P301+P312, P303+P361+P353, P304+P340, P305+P351+P338, P310, P370+P378, P403+P233, P405, P501

GHS pictograms: GHS02 (Flame), GHS05 (Corrosion), GHS06 (Skull and crossbones), GHS08 (Health hazard), GHS09 (Environment)

Other hazards: CRITICAL: Becomes shock-sensitive and may decompose explosively at concentrations above ~88 wt% (e.g. due to evaporative concentration in dead legs or during sampling). Self-Accelerating Decomposition Temperature (SADT) is concentration- and package-size-dependent — typically in the 60-80°C range for commercial 80% grade. Not PBT/vPvB.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Pure substance / mixture: Marketed as technical mixture

Cumene hydroperoxide: CAS 80-15-9 — typically 75-86 wt% (PPCL OXI product stream concentration to be confirmed against process design basis)

Cumene (carrier): CAS 98-82-8 — balance, typically 14-25 wt%

Note: Composition varies along the OXI/CDN train; actual stream assay should be obtained from PPCL process data sheets for node-specific HAZOP work

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove victim to fresh air immediately. If breathing is difficult, administer oxygen. If not breathing, give artificial respiration. Obtain medical attention immediately.

Skin contact: Remove contaminated clothing immediately. Flush skin with large amounts of water for at least 15-20 minutes. Do not rub. Obtain medical attention — treat as a chemical burn.

Eye contact: Flush eyes with copious amounts of water for at least 15-20 minutes, holding eyelids open. Remove contact lenses if present and easy to do. Obtain immediate medical/ophthalmological attention.

Ingestion: Do NOT induce vomiting. Rinse mouth with water. If conscious, give water to drink. Obtain immediate medical attention — risk of severe internal corrosive injury.

Notes to physician: Treat as corrosive chemical burn with possible systemic toxicity (hepatic/renal effects per H373). Monitor for delayed pulmonary effects after inhalation. No specific antidote.

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Water spray/fog (large quantities to cool and dilute), alcohol-resistant foam, dry chemical, carbon dioxide for small fires

Unsuitable extinguishing media: Do not use a solid water jet directly on hot/decomposing material — may cause violent splattering. Do not confine in closed spaces.

Special hazards: Thermally unstable organic peroxide — heating, fire exposure or contamination can trigger self-accelerating exothermic decomposition, generating flammable vapours (acetone, AMS) and oxygen, which can re-ignite or rupture/explode closed containers. Vapours are heavier than air and may travel to ignition sources.

Special protective equipment for firefighters: Full firefighting turnout gear with self-contained breathing apparatus (SCBA) in positive-pressure mode. Fight fire from maximum distance or use unmanned monitors/hose holders. Cool exposed containers with water from a safe distance; evacuate area if rising sound from venting safety devices is heard.

Hazardous combustion products: Carbon monoxide, carbon dioxide, phenol, acetone, alpha-methylstyrene vapours, oxygen

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Evacuate non-essential personnel. Eliminate all ignition sources. Wear full PPE (Section 8) including chemical-resistant suit, gloves, goggles and, for vapour/spray releases, SCBA. Avoid friction, shock and contamination of spilled material.

Environmental precautions: Prevent spilled material from entering drains, sewers, watercourses or soil. Bund/contain to limit spread.

Containment and cleanup: Absorb with inert, non-combustible absorbent (vermiculite, sand, diatomaceous earth) — do NOT use combustible absorbents such as sawdust or cellulose. Avoid sealing recovered material in air-tight containers (allow off-gassing); transfer to suitable labelled containers for controlled decomposition/disposal. Wash contaminated area with large amounts of water.

Reference to other sections: See Section 7 (Handling & Storage), Section 8 (PPE) and Section 13 (Disposal)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Handle in closed systems with adequate local exhaust ventilation. Avoid heat, sparks, open flame and other ignition sources. Avoid contact with skin, eyes and clothing. Avoid contamination with reducing agents, acids, bases, amines or transition-metal salts (Co, Mn, Fe, Cu) which catalyse decomposition. Bond/ground equipment during transfer; avoid friction and mechanical shock.

Storage conditions: Store in a cool (recommended <30°C, refrigerated for higher concentrations), well-ventilated, fire-resistant area away from direct sunlight.

Temperature limits: Maintain well below the SADT of the stored concentration (typically 60-80°C for ~80 wt% grade); continuous temperature monitoring with high-temperature alarm and emergency cooling/dump provisions recommended for CDN concentration vessels.

Pressure: Store in vented containers/tanks — do not seal; provide adequately sized pressure relief/emergency venting to accommodate decomposition gas generation.

Segregation requirements: Segregate from reducing agents, strong acids/bases, amines, heavy-metal compounds (Co, Mn, Cu, Fe salts), combustible materials and other incompatible organic peroxides.

Special requirements: Avoid copper and copper-alloy (brass/bronze) wetted components — metal ion contamination accelerates decomposition. Provide continuous temperature monitoring on CHP transfer lines and CDN concentration/decomposer vessels; design relief systems for runaway decomposition scenario per DIERS methodology.

Specific end-use recommendations: Process only as a dilute (≤ 80 wt%) solution in cumene; minimise hold-up time and avoid dead-legs where evaporative concentration above 88 wt% could occur.

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): No formally established ACGIH TLV, OSHA PEL or NIOSH REL for cumene hydroperoxide specifically. Where CHP is present with cumene, apply the cumene OEL (see SDS_98-82-8) as a surrogate for the carrier solvent. PPCL should establish an internal occupational exposure limit based on supplier guidance pending publication of an authoritative OEL.

TWA: Not established (see note above)

STEL: Not established

Ceiling: Not established

Skin notation: Yes — corrosive/absorbed through skin

IARC/NTP/OSHA carcinogen status: Not classified as a carcinogen by IARC, NTP or OSHA (insufficient data)

Engineering controls: Closed processing and transfer systems; local exhaust ventilation at sampling points and pump seals; continuous gas detection where vapour release is possible

PPE — Respiratory: Where engineering controls are insufficient, use a full-facepiece air-purifying respirator with organic vapour (OV) cartridge; SCBA for emergency response or high-concentration release

PPE — Gloves: Chemical-resistant gloves (e.g. butyl rubber, Viton, or laminate film); avoid nitrile for prolonged contact with concentrated CHP — verify breakthrough time with glove manufacturer

PPE — Eyes: Chemical splash goggles plus full face shield for transfer/sampling operations

PPE — Body: Chemical-resistant apron/suit; avoid materials containing copper-alloy fasteners in direct product contact areas

Biological monitoring: Not established

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless to pale yellow liquid (technical ~80% grade)

Odour: Characteristic sharp/acrid organic odour

Odour threshold: Not established

pH: Not applicable (organic peroxide; not typically aqueous)

Melting point / Freezing point: Approx. -5 to -15°C (technical grade; varies with cumene content)

Boiling point (at 1 atm): ~153°C (pure substance); technical solution decomposes before reaching atmospheric boiling point under prolonged heating

Flash point: ~79°C (closed cup, technical 80% solution); some pure-grade references report values as low as 57°C — use site-specific supplier SDS value for design

Evaporation rate: Low (low vapour pressure)

Flammability (solid/gas): Not applicable — combustible liquid (Flam. Liq. 4 / H227)

Flammability limits in air: Not well characterised for CHP itself; treat per cumene carrier solvent for vapour-phase flammability assessment (LEL ~0.8-0.9% v/v, UEL ~6.5% v/v for cumene)

Vapour pressure (at 20°C): <0.13 kPa (<1 mmHg) — low volatility

Vapour density (air = 1): ~5.2 (heavier than air — vapours will accumulate at low points)

Relative density (water = 1): ~1.02-1.06 (technical 80% solution)

Solubility (water): Slightly soluble (<1 g/L at 20°C); miscible with most organic solvents

Partition coefficient (log K_{ow}): ~2.2 (estimated)

Auto-ignition temperature: Not meaningfully established — material decomposes exothermically before sustained autoignition occurs

Decomposition temperature: CRITICAL FOR CHP — DSC exothermic onset reported from ~80°C for 88 wt% CHP in cumene; self-accelerating decomposition (SADT) typically occurs in the 60-80°C range depending on package size/concentration; violent/runaway decomposition of 91-95% material reported near 150°C. Heat of decomposition >1300

J/g for 88 wt% CHP. Decomposition is catalysed by acids, metal ions and contamination.

Viscosity: Low viscosity, similar to light oils (~few cP at 20°C)

Explosive properties: Not classified as an explosive under transport criteria for the ≤80-86% grade, but the pure material (>88 wt%) is shock-sensitive — Organic Peroxide Type E

Oxidising properties: Organic peroxide — acts as a strong oxidiser on decomposition

Molecular weight: 152.19 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Thermally unstable organic peroxide; the O-O bond decomposes exothermically, and the reaction can self-accelerate (autocatalytic) once initiated

Chemical stability: Stable when stored cool (well below SADT), in the dark, free of contamination, and in vented containers. Not stable indefinitely — degrades slowly even under recommended storage, generating trace decomposition products and pressure

Possibility of hazardous reactions: Runaway exothermic decomposition with rapid gas (oxygen) evolution and generation of flammable vapours; potential for vessel overpressure/rupture if relief is inadequate

Conditions to avoid: Heat above storage temperature limits, friction, mechanical shock, static discharge, exposure to light, evaporative concentration above ~88 wt%

Incompatible materials: Reducing agents, strong acids and bases, amines, transition-metal compounds (cobalt, manganese, iron, copper salts/oxides), combustible materials, other organic peroxides

Hazardous decomposition products: ACETONE, PHENOL, CUMENE, ALPHA-METHYLSTYRENE, ACETOPHENONE and OXYGEN, with significant heat release — this is the basis of the controlled CDN decomposition reaction but represents a major runaway hazard if uncontrolled (loss of cooling / acid catalyst maldistribution)

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation, dermal contact, eye contact, ingestion

Acute toxicity — Oral LD50: ~250-400 mg/kg, rat (literature range; classified Acute Tox. 4)

Acute toxicity — Dermal LD50: ~2,600-2,700 mg/kg, rabbit (classified Acute Tox. 4)

Acute toxicity — Inhalation LC50: Classified Acute Tox. 3 (inhalation) based on vapour/mist studies; specific LC50 not consistently reported in open literature — obtain from manufacturer SDS

Skin corrosion/irritation: Corrosive — causes severe skin burns (H314)

Serious eye damage/irritation: Causes serious eye damage (H314)

Respiratory/skin sensitisation: Not classified as a respiratory sensitiser; some reports of dermal sensitisation potential — use appropriate PPE

Germ cell mutagenicity: Not classified; limited in-vitro genotoxicity data reported, inconclusive overall

Carcinogenicity: Not classified by IARC, NTP, ACGIH or OSHA

Reproductive toxicity: No reliable data identified

STOT (single exposure): May cause respiratory irritation on inhalation

STOT (repeated exposure): H373 — may cause damage to kidneys and liver through prolonged or repeated exposure

Aspiration hazard: Not classified

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC50/EC50): Classified Aquatic Acute 2 / Aquatic Chronic 2 (H411 — toxic to aquatic life with long lasting effects); specific LC50/EC50 values not consistently published — obtain from manufacturer ecotoxicology data

Persistence and degradability: Expected to decompose relatively readily (peroxide instability); degradation products (phenol, acetone, AMS) have their own ecotoxicity profiles

Bioaccumulative potential: Low (estimated log Kow ~2.2)

Mobility in soil: Moderate; not expected to persist due to instability

Other adverse effects: No ozone-depletion or specific climate-change data; avoid release to the environment, including via firewater runoff

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of as hazardous chemical waste via a licensed waste contractor. Small laboratory quantities may be diluted and catalytically decomposed (e.g. with dilute ferrous sulfate/sodium metabisulfite) before disposal — only by trained personnel under controlled conditions. Do not discharge to drains.

Contaminated packaging: Triple-rinse with compatible solvent under controlled conditions; dispose of as hazardous waste; do not re-use empty containers

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535 / Factory Act waste management requirements; Basel Convention for any transboundary movement

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 3105 (or UN 3107, depending on formulation/concentration — confirm against current transport classification for the specific commercial grade)

Proper shipping name: ORGANIC PEROXIDE TYPE E, LIQUID (Cumene hydroperoxide, ≤86%, in cumene) — confirm exact entry with current Dangerous Goods regulations

Transport hazard class(es): Class 5.2 (Organic peroxide), with subsidiary risks of Class 6.1 (Toxic) and/or Class 8 (Corrosive) depending on grade

Packing group: II (typical for this entry)

Environmental hazards (marine pollutant): Yes — marine pollutant (Aquatic Chronic 2)

Special precautions for user: Temperature-controlled transport may be required if SADT of the package is below ambient/transport temperatures

Bulk transport (MARPOL/IBC): If transported in bulk by sea, classify per IBC Code; verify entry with shipper's dangerous goods declaration

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

Thailand — additional note: Organic peroxides are subject to additional storage/quantity controls under the Factories Act and Hazardous Substances Act — confirm permit category for CDN inventory with PPCL regulatory affairs.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 80-15-9)

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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